MERS-CoV Surveillance Gaps Analysis: Lessons from South Batinah, Oman

Introduction

The delayed detection of the first confirmed Middle East Respiratory Syndrome Coronavirus (MERS-CoV) case in South Batinah, Oman, revealed important weaknesses in the local public health surveillance system. The patient, a 54-year-old man with hypertension and diabetes, developed fever and respiratory symptoms around August 9, 2017. Although he sought care at multiple healthcare facilities, MERS-CoV was not confirmed until August 28.

The delay created several missed opportunities for early isolation, diagnostic testing, contact identification, and infection-control intervention. During the period before diagnosis, the patient remained socially active as a mosque imam and participated in religious activities and gatherings involving individuals from Oman, Saudi Arabia, and the United Arab Emirates. His movement through primary, private, regional, and tertiary healthcare facilities also created the potential for healthcare-associated exposure.

This report evaluates the clinical and operational factors that contributed to delayed detection, examines the consequences for the public health response, and provides recommendations for strengthening surveillance and infection-control systems.

Objectives

The objectives of this analysis were to:

1. Reconstruct the progression of the case from symptom onset through laboratory confirmation.
2. Assess whether the patient met Oman’s surveillance criteria for severe respiratory illness.
3. Identify gaps in case recognition, triage, reporting, testing, infection control, and contact monitoring.
4. Evaluate the public health effects of delayed detection.
5. Recommend practical strategies for improving future surveillance and outbreak response.

Methods

The case was reviewed using an outbreak-investigation and surveillance-system assessment approach. Clinical and epidemiologic information was organized across five major areas: case detection, triage and referral, laboratory testing and notification, infection prevention and control, and contact investigation.

The patient's symptoms, underlying conditions, healthcare visits, hospital transfers, social exposures, and clinical progression were compared with Oman's Intensified Surveillance of Severe Acute Respiratory Infections, or SARI-IS, criteria. The analysis also considered the sensitivity and specificity of the surveillance definitions and whether healthcare workers followed established testing and notification procedures.

Because this report is based on a teaching case developed from a 2017 field investigation, it does not contain independently collected patient data. The available materials also do not provide the final outcomes of every contact or confirm that secondary transmission occurred.

Case Overview and Timeline

The patient began experiencing fever and respiratory symptoms around August 9, 2017. During the early stage of his illness, he continued participating in religious and social activities and interacted with hundreds of people.

On August 21, he visited a primary health center with respiratory symptoms and diarrhea. Testing identified *Entamoeba histolytica*, and his erythrocyte sedimentation rate was elevated at 45 mm/hr. He was prescribed metronidazole, Augmentin, and amoxicillin and discharged without additional respiratory evaluation or referral.

On August 23, the patient was transferred from a private hospital to the emergency department of a regional hospital. He presented with a fever above 38°C, productive cough, shortness of breath, severe pneumonia, and respiratory distress. Despite these symptoms, the initial investigation focused on tuberculosis and other viral or bacterial infections without including MERS-CoV.

His condition deteriorated the following day, requiring admission to the intensive care unit, intubation, and mechanical ventilation. On August 28, he was transferred to Sultan Qaboos University Hospital in Muscat, where testing confirmed MERS-CoV. The regional epidemiologist was notified on August 29.

Approximately 19 days passed between the reported onset of symptoms and laboratory confirmation.

Findings

Case Recognition and Diagnostic Anchoring

The patient had several clinical characteristics that should have raised suspicion for an emerging respiratory infection. These included fever, productive cough, shortness of breath, pneumonia, severe respiratory distress, rapid clinical deterioration, ICU admission, and mechanical ventilation.

However, the initial diagnostic assessment focused primarily on pneumonia and tuberculosis. Clinicians also placed considerable emphasis on the absence of reported camel exposure. This narrowed the assessment of potential MERS-CoV risk and overlooked other relevant information, including the patient's severe clinical presentation, chronic health conditions, attendance at large gatherings, and interactions with visitors from areas where MERS-CoV was circulating.

The case demonstrates how relying too heavily on one recognized exposure can reduce surveillance sensitivity. Exposure history should inform risk assessment, but the absence of a known exposure should not prevent testing when the clinical presentation meets severe respiratory surveillance criteria.

Triage and Referral

The patient passed through several healthcare settings before receiving an appropriate diagnosis. At the primary health center, the focus on gastrointestinal findings may have distracted clinicians from the seriousness of his respiratory symptoms.

The subsequent investigation found that the health center experienced a high volume of patients, heavy staff workloads, and the absence of a structured triage system. These operational pressures contributed to an incomplete assessment and failure to refer the patient for further respiratory evaluation.

Transitions between the primary health center, private hospital, regional hospital, and tertiary hospital also increased the possibility that important clinical and epidemiologic information would be lost. A standardized respiratory-illness alert should have followed the patient across each level of care.

SARI-IS Case Definition

Oman introduced Intensified Surveillance of Severe Acute Respiratory Infections in hospitals in November 2016. The SARI-IS definition included admitted patients with fever above 38°C and cough or breathing difficulty, along with one or more of the following:

- Evidence of severe illness progression
- Pneumonia or acute respiratory distress syndrome
- ICU admission or mechanical ventilation
- No alternative diagnosis within 72 hours
- Chronic health conditions or other high-risk characteristics
- Relevant travel, healthcare, animal, cluster, or confirmed-case exposure

The patient met several of these criteria. He had fever, cough, breathing difficulty, severe pneumonia, ICU admission, mechanical ventilation, diabetes, hypertension, and no immediate alternative diagnosis that fully explained his illness.

Despite meeting the criteria, he was not initially tested for MERS-CoV. This indicates that the primary problem was not the absence of a surveillance definition but inconsistent implementation of the existing definition.

Testing and Notification

The SARI-IS program required qualifying ICU respiratory cases to be tested using real-time RT-PCR for MERS-CoV and other respiratory pathogens. However, MERS-CoV was excluded from the patient's initial diagnostic testing.

The investigation found that some physicians were not fully familiar with the SARI-IS definition, MERS-CoV case criteria, or the notification process. Staff had not received sufficient follow-up training after the program was introduced. Some clinicians also expressed frustration with notification requirements and perceived delays in receiving laboratory results.

These findings suggest that surveillance procedures were not adequately integrated into routine clinical workflows. A policy is unlikely to improve detection unless clinicians understand when it applies, can initiate testing easily, and know exactly whom to notify.

Infection Prevention and Control

Infection-control procedures were not implemented consistently during the patient's early healthcare encounters. Because MERS-CoV was not initially suspected, the patient was not promptly isolated under the appropriate respiratory precautions.

The investigation identified healthcare workers who had participated in aerosol-generating procedures or had close contact with the patient without adequate personal protective equipment. Several high-risk contacts had not been reached when the investigation team reviewed the hospital's response.

An emergency department physician who initially evaluated the patient also continued working despite experiencing respiratory symptoms. Allowing a symptomatic exposed healthcare worker to remain on duty created an additional risk of healthcare-associated transmission.

Contact Investigation

The investigation identified 73 healthcare worker contacts across the healthcare facilities involved in the patient's care. Of these workers, 20 were classified as high risk because they had unprotected close contact with the patient or participated in aerosol-generating procedures.

Investigators also identified 36 family contacts, including three who were symptomatic. Symptomatic contacts required immediate clinical assessment and testing, while other contacts required daily symptom monitoring for 14 days following their last exposure.

Community contact tracing was more challenging. As a mosque imam, the patient had organized religious activities and attended gatherings involving hundreds of people. The time that passed before diagnosis made it difficult to identify every person he encountered and determine the likely source of his infection.

The available materials do not establish that secondary transmission occurred. Nevertheless, the number and variety of contacts illustrate how delayed detection can substantially increase the complexity and cost of an outbreak response.

Major Surveillance Gaps

The analysis identified six interconnected weaknesses.

1. Limited Case Recognition

Clinicians focused heavily on camel exposure and alternative diagnoses, even though the patient's clinical severity met intensified surveillance criteria.

2. Inadequate Triage and Escalation

The absence of a consistent triage system, combined with high workloads, contributed to delayed referral and incomplete assessment.

3. Delayed Diagnostic Testing

MERS-CoV was not included in the initial testing despite the patient's severe respiratory symptoms and ICU admission.

4. Unclear Notification Procedures

Some healthcare workers were unfamiliar with surveillance-reporting requirements and the appropriate notification pathway.

5. Inconsistent Infection-Control Practices

The patient was not isolated early, PPE use was inconsistent, and exposed symptomatic staff were not immediately removed from duty.

6. Incomplete Contact Follow-Up

Some high-risk healthcare contacts were not promptly notified or monitored, increasing the possibility that symptoms or additional cases could be identified late.

Discussion

This case demonstrates the difference between having a surveillance policy and having a surveillance system that works consistently in practice. Oman had already established SARI-IS guidance, but the guidance was not reliably translated into clinical decisions.

The surveillance failure resulted from a sequence of connected problems rather than one isolated diagnostic mistake. Workload pressures affected triage. Narrow assumptions affected clinical recognition. Limited training affected testing and notification. Delayed diagnosis affected infection-control practices and expanded the number of contacts requiring investigation.

The case also illustrates the balance between sensitivity and specificity. A narrow exposure-based definition may reduce unnecessary testing, but it can miss patients whose exposure is unknown or atypical. In an area where MERS-CoV is considered endemic, unexplained severe respiratory illness should be sufficient to trigger initial isolation and testing.

A broader first-stage screen would not eliminate specificity. Clinical criteria could initiate precautions and laboratory testing, while epidemiologic investigation and test results would determine whether the patient should remain classified as a suspected or confirmed case.

Public Health Impact

Increased Exposure Risk

The patient remained active in healthcare, household, religious, and social environments before MERS-CoV was confirmed. This expanded the number of people who required assessment and monitoring.

Greater Contact-Tracing Burden

Investigators had to reconstruct interactions across several healthcare facilities and community settings. The delay made it more difficult to locate contacts and identify the probable source of infection.

Strain on Healthcare Resources

The response required contact classification, staff monitoring, diagnostic testing, infection-control review, risk communication, and coordination among multiple healthcare and public health agencies.

Risk of Healthcare-Associated Transmission

Delayed isolation, inconsistent PPE use, aerosol-generating procedures, and continued work by a symptomatic exposed physician increased the potential for transmission within healthcare settings.

Reduced Public Confidence

Delayed detection can weaken confidence in the health system, particularly when the public learns that an infected person interacted with many individuals before being diagnosed. Timely and transparent communication is necessary to prevent misinformation and unnecessary fear.

Recommendations

1. Standardize Early Recognition

Healthcare facilities should use a two-stage screening process. The first stage should identify severe or unexplained respiratory illness using clinical criteria. The second stage should assess travel, healthcare exposure, contact with confirmed cases, animal exposure, clusters, and attendance at large gatherings.

Meeting the severe clinical criteria should be sufficient to begin isolation, testing, and notification, even when a traditional exposure is not identified.

2. Strengthen Triage and Referral

Primary care centers and emergency departments should establish clear escalation procedures for patients with fever, respiratory distress, unexplained pneumonia, rapid deterioration, high-risk health conditions, or abnormal findings suggesting systemic illness.

Staff should understand when a patient requires additional evaluation, where the patient should be referred, and how relevant risk information should be communicated to the receiving facility.

3. Provide Recurring Workforce Training

Training should occur regularly rather than only when a surveillance program is introduced. Scenario-based exercises should cover:

- MERS-CoV and SARI-IS case definitions
- Diagnostic testing requirements
- Specimen collection and transportation
- Notification procedures
- PPE selection and use
- Aerosol-generating procedures
- Contact-risk classification
- Work restrictions for exposed staff

4. Improve Diagnostic Access and Turnaround

Standing testing orders should be established for ICU patients who meet severe respiratory criteria. Health systems should define expected laboratory turnaround times and create automatic alerts for positive or concerning results.

Where feasible, testing access or specimen transportation should be decentralized so that regional facilities are not dependent on delayed tertiary-level processes.

5. Standardize Infection-Control Practices

Patients who meet severe respiratory triggers should be isolated immediately while diagnostic testing is completed. Healthcare facilities should routinely audit PPE availability, staff compliance, respiratory-isolation procedures, and practices during aerosol-generating procedures.

Exposed healthcare workers who develop symptoms should be removed from duty and evaluated promptly.

6. Establish Closed-Loop Contact Monitoring

A centralized contact registry should document:

- Contact information
- Exposure category
- Date of last exposure

- Daily symptom status
- Testing status and results
- Assigned monitor
- Final disposition

Missed follow-up attempts and newly reported symptoms should generate automatic escalation alerts.

7. Improve Information Integration

Facility notifications, laboratory results, patient transfers, and contact-monitoring information should be connected through a shared reporting system. Local, regional, and national teams should have access to the same current information.

Electronic alerts can improve timeliness, but each alert must still have an identified person responsible for reviewing it and documenting the resulting action.

8. Strengthen Risk Communication

Healthcare workers, contacts, and community members should receive timely and accurate information. Communications should distinguish confirmed findings from potential risks, explain monitoring recommendations clearly, and avoid unnecessary fear or stigma.

Conclusion

The first confirmed MERS-CoV case in South Batinah was detected only after multiple healthcare encounters and significant clinical deterioration. Although the patient met Oman's SARI-IS criteria, the regional surveillance pathway did not activate when it should have.

The delay resulted from connected weaknesses in triage, case recognition, diagnostic testing, reporting, infection control, and contact follow-up. These weaknesses extended the period of potential exposure and increased the demands placed on the public health response.

The central lesson from this case is that timely surveillance depends on the reliability of the entire detection-to-response process. Case definitions must be clinically usable, healthcare workers must receive recurring training, reporting and laboratory procedures must be clear, and every surveillance alert must lead to a documented action. Strengthening these areas would improve Oman's ability to detect and respond to future emerging respiratory threats.

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